

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q20-1548

Speed 138 kV
October 27, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1548. The Interconnection Customer proposes a 673.7 MW natural gas combined cycle facility interconnecting at the Speed 138 kV substation in the Kentucky area. The request was received on August 18, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 355 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 14 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$55,765,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Oates Reit Substation - 56Th T 138kV
Contingency: Speed - Silver Bay Power 138kV
Loading: 324.8% (Rating: 247 MVA)
DFAX: 100.0%
Pre-project loading: 51.9%
2. Facility: 56Th St - Santa Monia 138kV
Contingency: Silver Bay Power - Speed 138kV line
Loading: 280.6% (Rating: 298 MVA)
DFAX: 100.0%
Pre-project loading: 54.5%
3. Facility: Brockport - Narod 138kV
Contingency: Silver Bay Power - Speed 138kV
Loading: 277.7% (Rating: 305 MVA)
DFAX: 100.0%
Pre-project loading: 56.9%
4. Facility: Santa Monica - Hambrick 138kV
Contingency: Speed - Silver Bay Pwer 138kV
Loading: 267.0% (Rating: 321 MVA)
DFAX: 100.0%
Pre-project loading: 57.0%
5. Facility: Barnham - Narod 138kV
Contingency: Speed Substation - Silver Bay Power 138kV line
Loading: 256.4% (Rating: 344 MVA)
DFAX: 100.0%
Pre-project loading: 60.5%

6. Facility: Gaga - Barnham 138kV
Contingency: Speed - Silver Bay Power 138kV
Loading: 246.1% (Rating: 367 MVA)
DFAX: 100.0%
Pre-project loading: 62.3%
7. Facility: Silver Bay Power - Speed 138kV Line
Contingency: Speed - Gaga 138kV
Loading: 237.3% (Rating: 388 MVA)
DFAX: 100.0%
Pre-project loading: 63.8%
8. Facility: Speed - Gaga 138kV
Contingency: Speed - Silver Bay Power 138kV
Loading: 237.3% (Rating: 388 MVA)
DFAX: 100.0%
Pre-project loading: 63.8%
9. Facility: Silver Bay Power - Hotel Rd 138kV
Contingency: Speed - Gaga 138kV
Loading: 182.8% (Rating: 208 MVA)
DFAX: 31.8%
Pre-project loading: 79.7%
10. Facility: Silver Bay Power - Damon 138kV
Contingency: Hotel Road - Silver Bay Power 138kV
Loading: 135.3% (Rating: 230 MVA)
DFAX: 33.9%
Pre-project loading: 36.2%
11. Facility: Cleveland - W Mesquite Substation 138kV Circuit 1
Contingency: West Mesquite - Cleveland 138kV Line
Loading: 107.0% (Rating: 282 MVA)
DFAX: 7.3%
Pre-project loading: 89.6%
12. Facility: Cleveland - W Mesquite 138kV
Contingency: Cleveland - W Mesquite 138kV Ckt 1
Loading: 107.0% (Rating: 282 MVA)
DFAX: 7.3%
Pre-project loading: 89.6%
13. Facility: Murphys Substation - Stelco 138kV
Contingency: Pine Bluff Watson Chapel - Carraizo 1201 Substation 345kV Line
Loading: 106.4% (Rating: 294 MVA)
DFAX: 14.9%
Pre-project loading: 72.2%
14. Facility: Dewitt - Stelco 138kV
Contingency: Carraizo 1201 - Pine Bluff Watson Chapel 345kV
Loading: 104.4% (Rating: 204 MVA)
DFAX: 7.8%
Pre-project loading: 78.5%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Oates Reit Substation - 56Th T 138kV	Rebuild line with higher-capacity conductor	\$2,226,000	72% (\$1,594,000)
56Th St - Santa Monia 138kV	Rebuild line with higher-capacity conductor	\$2,615,000	72% (\$1,873,000)
Brockport - Narod 138kV	Rebuild line with higher-capacity conductor	\$3,114,000	72% (\$2,231,000)
Santa Monica - Hambrick 138kV L	Rebuild line with higher-capacity conductor	\$4,974,000	72% (\$3,563,000)
Barnham - Narod 138kV	Rebuild line with higher-capacity conductor	\$20,222,000	72% (\$14,484,000)
Gaga - Barnham 138kV	Rebuild line with higher-capacity conductor	\$2,407,000	72% (\$1,724,000)
Silver Bay Power - Speed 138kV Line	Rebuild line with higher-capacity conductor	\$18,877,000	72% (\$13,520,000)
Speed Substation - Gaga 138kV line	Rebuild line with higher-capacity conductor	\$3,310,000	72% (\$2,371,000)
Silver Bay Power - Hotel Rd 138kV	Rebuild line with higher-capacity conductor	\$17,825,000	56% (\$10,036,000)
Silver Bay Power - Damon 138kV	Rebuild line with higher-capacity conductor	\$2,598,000	72% (\$1,861,000)
Cleveland - W Mesquite Substation 138kV Circuit 1	Reconductor line	\$758,000	26% (\$198,000)
W Mesquite - Cleveland 138kV line	Reconductor line	\$737,000	26% (\$193,000)
Murphys Substation - Stelco 138kV	Reconductor line	\$2,395,000	35% (\$831,000)
Dewitt - Stelco Substation 138kV	Reconductor line	\$4,063,000	32% (\$1,286,000)
	Total allocated to project		\$55,765,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.